WORKED SOLUTION

STEP II 2016, Question 5

Pure • Binomial Coefficient Identities

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q5

The key idea

All three identities come from one trick: write a function as a product in two ways and compare the coefficient of a single power of x. A sum of products of binomial coefficients, with indices adding to a fixed total, is exactly what you get when you multiply two power series and read off one coefficient. The question's convention that $\binom{p}{q} = 0$ outside $p \geq q \geq 0$ is what lets the sums run over a tidy range of j without fuss about which terms exist.

Solution

Part (i)

For $|x| < 1$ the binomial series gives the coefficient of x^n in $(1-x)^{-N}$ as

$$\frac{(-N)(-N-1)\cdots(-N-n+1)}{n!}(-1)^n = \frac{N(N+1)\cdots(N+n-1)}{n!}$$
$$= \frac{(N+n-1)!}{n!\,(N-1)!} = \binom{N+n-1}{n}.$$

The n minus signs from the factors cancel the n minus signs from $(-x)^n$. So

$$(1-x)^{-N} = \sum_{n=0}^{\infty} \binom{N+n-1}{n} x^n \qquad (|x| < 1).$$

(When $n = 0$ the coefficient is $\binom{N-1}{0} = 1$. This is fine because $N \geq 1$.)

Now $(1-x)^{-1}(1-x)^{-N} = (1-x)^{-(N+1)}$. By the expansion above with $N+1$ in place of N, the coefficient of x^n on the right is $\binom{N+n}{n}$.

On the left, $(1-x)^{-1} = \sum_{i \geq 0} x^i$ has every coefficient equal to 1. A term x^j from $(1-x)^{-N}$ must pair with x^{n-j} from $(1-x)^{-1}$, and j runs from 0 to n. So the coefficient

of x^n in the product is

$$\sum_{j=0}^{n} 1 \cdot \binom{N+j-1}{j}.$$

Both power series converge for $|x| < 1$ and represent the same function there, so their coefficients agree. That gives

$$\sum_{j=0}^{n} \binom{N+j-1}{j} = \binom{N+n}{n},$$

as required.

Part (ii)

Use $(1+x)^{m+n} = (1+x)^m (1+x)^n$, where all three are polynomials. With the question's convention we can write

$$(1+x)^m = \sum_{j \geq 0} \binom{m}{j} x^j, \qquad (1+x)^n = \sum_{k \geq 0} \binom{n}{k} x^k,$$

since the coefficients with $j > m$ or $k > n$ are zero. The coefficient of x^r in the product comes from the pairs with $j + k = r$, that is $k = r - j$ with $0 \leq j \leq r$:

$$\sum_{j=0}^{r} \binom{m}{j} \binom{n}{r-j}.$$

Ask yourself What about the terms where $j > m$ or $r - j > n$? Those powers don't appear in the polynomials.

Those terms contain a factor $\binom{m}{j}$ with $j > m$ or $\binom{n}{r-j}$ with $r - j > n$, so they are zero by the convention. They contribute nothing, and they also correspond to no actual product of terms, so the sum over $0 \leq j \leq r$ is exactly right. The coefficient of x^r in $(1+x)^{m+n}$ is $\binom{m+n}{r}$. Polynomials that are equal have equal coefficients, so

$$\binom{m+n}{r} = \sum_{j=0}^{r} \binom{m}{j} \binom{n}{r-j}.$$

Part (iii)

The factor $(-1)^j \binom{m+j-1}{j}$ suggests part (i) with $-x$ in place of x. Replacing x by $-x$ in the expansion from part (i), with m in place of N, gives

$$(1+x)^{-m} = \sum_{j=0}^{\infty} (-1)^j \binom{m+j-1}{j} x^j \qquad (|x| < 1).$$

Ask yourself Which product should we use?

The other factor $\binom{N+m}{n-j}$ is a coefficient of $(1+x)^{N+m}$, and the answer $\binom{N}{n}$ is a coefficient of $(1+x)^N$. Since m is a positive integer, for $|x|<1$ we have

$$(1+x)^{N+m}(1+x)^{-m} = (1+x)^N.$$

On the left, $(1+x)^{N+m} = \sum_{k\geq 0}\binom{N+m}{k}x^k$ (using the convention for $k > N+m$). The coefficient of x^n comes from x^j in $(1+x)^{-m}$ times x^{n-j} in $(1+x)^{N+m}$, for $0 \leq j \leq n$:

$$\sum_{j=0}^{n}\binom{N+m}{n-j}(-1)^j\binom{m+j-1}{j}.$$

On the right, the coefficient of x^n in $(1+x)^N$ is $\binom{N}{n}$. When $n > N$ this is 0 by the convention, which is also correct, because $(1+x)^N$ has degree N. The two sides are equal on $|x|<1$, so their power series coefficients agree:

$$\sum_{j=0}^{n}(-1)^j\binom{N+m}{n-j}\binom{m+j-1}{j} = \binom{N}{n}.$$

Checks

- Part (i) with $N=2$, $n=3$: $1+2+3+4 = 10 = \binom{5}{3}$.
- Part (ii) with $m=n=2$, $r=2$: $1\cdot 1 + 2\cdot 2 + 1\cdot 1 = 6 = \binom{4}{2}$.
- Part (iii) with $m=2$, $N=3$, $n=2$: $\binom{5}{2}\binom{1}{0} - \binom{5}{1}\binom{2}{1} + \binom{5}{0}\binom{3}{2} = 10 - 10 + 3 = 3 = \binom{3}{2}$. With $n > N$ the alternating sum should come out as 0. For example, $m=1$, $N=1$, $n=2$ gives $\binom{2}{2} - \binom{2}{1} + \binom{2}{0} = 1 - 2 + 1 = 0$.

All three identities were also confirmed by computer for every $m, N \leq 9$ and $n \leq 24$.

Common slips

- Losing track of the signs in the coefficient of $(1-x)^{-N}$. There are two lots of $(-1)^n$, and they cancel.
- In part (iii), forgetting that the series for $(1+x)^{-m}$ is infinite and holds only for $|x|<1$. Comparing coefficients is valid because two power series that agree on an interval have the same coefficients.

Beyond STEP

This is the method of *generating functions*: encode a sequence as the coefficients of a power series, and multiplying series turns into the convolution sums seen here. Part (ii)

is Vandermonde's identity, and part (i) is often called the hockey-stick identity. Both also have neat counting proofs, which are worth finding. Wilf's *generatingfunctionology* (free online) and Graham, Knuth and Patashnik, *Concrete Mathematics*, develop this toolkit much further.

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